Math 142, F20 **Name:** Answer
Quiz 4

<u>Instructions:</u> **You must show all of your work to receive credits for correct answers.**

Problem 1. (5 points) Evaluate $\int \frac{x-13}{x^2-2x-15}\,dx$.

$$\frac{x-13}{x^2-2x-15} = \frac{x-13}{(x-5)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3} = \frac{A(x+3)+B(x-5)}{(x-5)(x+3)}$$

Two roots $x=5, x=-3$

$\Rightarrow x-13 = A(x+3)+B(x-5)$ — ①

Let $x=5$, we get from ①: $5-13 = A(5+3)+B\cdot 0$

$-8 = 8A \Rightarrow A=-1$

Heaviside method

$x=-3$, we get from ①: $-3-13 = A\cdot 0 + B(-3-5)$

$-16 = -8B \Rightarrow B=2$

Thus $\int \frac{x-13}{x^2-2x-15}\,dx = \int \frac{-1}{x-5} + \frac{2}{x+3}\,dx$

$= -\ln|x-5| + 2\ln|x+3| + C$

Problem 2. (5 points) Evaluate $\int \frac{3x^2+x+4}{x^3+x}\,dx$

$$\frac{3x^2+x+4}{x^3+x} = \frac{3x^2+x+4}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{A(x^2+1)+x(Bx+C)}{x(x^2+1)}$$

$$= \frac{Ax^2+A+Bx^2+Cx}{x(x^2+1)} = \frac{(A+B)x^2+Cx+A}{x(x^2+1)}$$

$\Rightarrow \begin{cases} A+B=3 \\ C=1 \\ A=4 \end{cases} \Rightarrow A=4, B=-1, C=1$

Thus $\int \frac{3x^2+x+4}{x^3+x}\,dx = \int \frac{4}{x} + \frac{-x+1}{x^2+1}\,dx$.

$= \int \frac{4}{x} - \frac{x}{x^2+1} + \frac{1}{x^2+1}\,dx$

$= 4\ln|x| - \frac{1}{2}\ln(x^2+1) + \arctan x + C$.